The Chiral Completion Trilemma for Möbius-Incidence Atoms: Schatten-3 Motion versus Metric Collapse

Anonymous Authors

August 14, 2026

Abstract

We study a canonical chiral completion of source-derived Möbius incidence idempotents on a weighted divisibility Hilbert space. The oblique atom family is boundedly similar to coordinate projections, so its Dirichlet operator T_s belongs to $\mathcal{S}_q$ exactly when $q\Re s > 1$. For the holomorphic reflected block $\mathcal{B}_s = \left(\begin{smallmatrix} 0 & T_s \\ T^\sharp_{1-s} & 0 \end{smallmatrix}\right)$, the two legs give the sharp common strip $1/q < \Re s < 1 - 1/q$; hence $q = 3$ is the first integer order reaching the critical line. There $\mathcal{B}_{1/2+it}$ is compact self-adjoint but not Hilbert–Schmidt. We compute the native mixed Gram matrix exactly and prove that the fourth spectral moment has a positive isolated Fourier coefficient $4G^2_{pq}/(pq)$, so the self-adjoint spectrum genuinely moves with t. This motion is not arithmetically selective: exact mutated, composite-only, and generic poset controls reproduce it. Conversely, every bounded positive metric making the active idempotents self-adjoint has atom-diagonal conjugate Z^*GZ, forcing independent coordinate blocks and erasing all t-motion. Thus the canonical completions exhibit a precise trilemma between native motion, arithmetic selectivity, and metric orthogonality. The resulting third-regularized determinant is honest, but the construction is a t-dependent family rather than a fixed Hilbert–Pólya operator.

Research status. Candidate SD-C30 gives an honest third-regularized Fredholm determinant for a holomorphic reflected incidence family on $1/3 < \Re s < 2/3$. On $s = 1/2 + it$ the family is compact self-adjoint and its fourth moment moves exactly with t, but the operator itself depends on t and the same motion occurs on non-arithmetic controls. Every positive common orthogonalizing metric collapses the active family to independent atom coordinates and erases the motion. The strict tuple is `(A0_STRUCTURAL_ARITHMETIC_RELATION, A1_FAIL, A2_ANALYTIC_DETERMINANT, A3_FAIL, A4_FAIL)`. Hence `ROUTE_A_REJECTED`. Route B is locked and no target-zero data are used.

1 Introduction

Incidence inversion offers a particularly transparent route from a locally finite relation to a family of primitive idempotents. For divisibility, the zeta matrix Z and its Möbius inverse M produce

$$q_n = ZE_nM, \qquad q_nq_m = \delta_{nm}q_n.$$

The covers of the bottom element are the primes, so the associated Dirichlet family

$$T_s = \sum_p p^{-s} q_p$$

has an honest Euler determinant in its trace-class half-plane. The same construction exposes its apparent weakness: the q_p are oblique and globally similar to coordinate projections.

This paper asks whether the obliqueness can be turned into an asset. We join T_s to a holomorphically reflected adjoint leg and study

$$\mathcal{B}_s = \begin{pmatrix} 0 & T_s \\ T^\sharp_{1-s} & 0 \end{pmatrix}.$$

The reflection is complex-linear away from the critical line and becomes the Hilbert adjoint on it. The resulting answer has a positive and a negative half.

Positive half. The block belongs to $\mathcal{S}_q$ exactly on

$$\frac{1}{q} < \Re s < 1 - \frac{1}{q}.$$

Thus $\mathcal{S}_3$ is the first integer Schatten ideal whose common strip contains $\Re s = 1/2$. On that line the block is compact self-adjoint. Although the second trace is divergent, the fourth trace is honest, and unique factorization isolates a strictly positive Fourier coefficient. The native self-adjoint spectrum therefore really moves with t.

Negative half. The motion is carried by mixed Hilbert–Schmidt Gram coefficients of the oblique idempotents. Exact computations reproduce it after mutating divisibility, replacing the prime atoms by composites, and replacing divisibility by a seeded finite DAG. The effect is therefore generic incidence geometry rather than an arithmetic selector. Moreover, every positive metric that makes the active idempotents self-adjoint is classified by their coordinate commutant. It forces the active family to become independent one-dimensional atom blocks, whose reflected eigenvalues $\pm p^{-1/2}$ do not move with t.

Contributions. Our main results are:

1. the exact Schatten threshold and the minimal $\mathcal{S}_3$ critical strip;
2. closed formulas for the diagonal and mixed divisibility Gram coefficients;
3. a positive, isolated fourth-moment frequency proving genuine spectral motion after the $\det_3$ deletion;
4. a complete classification of bounded positive common orthogonalizing metrics on the active sector;
5. exact adversarial controls proving that native motion lacks arithmetic selectivity.

The result is deliberately not presented as a Hilbert–Pólya construction. The operator $\mathcal{B}_{1/2+it}$ depends on t; its third-regularized determinant is different from the original Euler determinant; and no target-zero information enters any definition or test. The strict route record is rejected.

Organization. Section 2 fixes the incidence and literature boundary. Section 3 proves the Schatten threshold. Section 4 derives the native Gram geometry. Section 5 establishes the first honest spectral motion. Section 6 proves metric rigidity. Section 7 reports the exact adversaries, and Section 8 states the completion trilemma and route decision.

2 Source compiler and prior boundary

2.1 Weighted divisibility realization

Fix $\eta > 1$ and let

$$\mathcal{H}_\eta = \left\{ x = (x_n)_{n\geq 1} : \sum_{n\geq 1} n^{2\eta} |x_n|^2 < \infty \right\}.$$

The divisibility zeta operator and its Möbius inverse are denoted by Z and $M = Z^{-1}$. In this range both are bounded and boundedly invertible. Let E_n be the coordinate projection and define

$$q_n = Z E_n M.$$

Then

$$q_n q_m = \delta_{nm} q_n, \qquad \sum_n q_n = I$$

in finite cutoffs, and the individual q_n extend as bounded rank-one idempotents in the countable realization.

For the atom set A of covers of the bottom element, define

$$D_s^A = \sum_{p \in A} p^{-s} E_p, \qquad T_s = Z D_s^A M = \sum_{p \in A} p^{-s} q_p.$$

In the unmutated divisibility relation, $A = \mathbb{P}$.

2.2 Holomorphic reflection

Let J be coordinate conjugation after normalizing the weighted coordinate basis. For source-real operators put

$$X^\sharp = J X^* J.$$

This reflection is complex-linear, reverses products, and preserves all Schatten norms. In unnormalized coordinates it is

$$X^\sharp = W_\eta^{-1} X^\top W_\eta, \qquad W_\eta = \operatorname{diag}(n^{2\eta}).$$

The reflected block

$$\mathcal{B}_s = \begin{pmatrix} 0 & T_s \\ T_{1-s}^\sharp & 0 \end{pmatrix}$$

is therefore holomorphic on every common Schatten domain. When $s = 1/2 + it$, the reflected leg equals T_s^*.

2.3 Marker ownership

The critical theorem in this paper is the arithmetic specialization $u = 1$. The inherited digit marker could be retained formally as

$$T_s(u) = \sum_p u^{\ell(p)} p^{-s} q_p.$$

Two firewalls are essential. First, the r-fold atom contribution is $u^{r\ell(p)}$, not $u^{\ell(p)}$. Second, when $|u| < 1$ the Schatten sum becomes

$$\sum_p |u|^{q\ell(p)} p^{-q\Re s},$$

so the threshold changes. This is a different regularized family, not analytic continuation of the $u = 1$ critical object.

2.4 Literature boundary

Rota's incidence-algebra framework supplies the Möbius source (Rota, 1964). Oblique projections and biorthogonal Riesz systems are classical operator geometry (Tang, 2000), and compatibility with positive diagonal weights has been studied in the weighted projection setting (Antezana et al., 2004). The square-root metric transfer is the operator form of symmetric orthogonalization (Löwdin, 1950).

Schatten ideals and modified determinants are standard (Simon, 2005). Higher determinant product formulas require explicit correction terms (Britz et al., 2020; Koutsonikos-Kouloumpis and Lesch, 2022); we therefore keep a power-by-power deletion ledger rather than treating $\det_3$

as an ordinary determinant. Recent work on Möbius inversion over locally finite posets remains combinatorial rather than chiral-operator theoretic (Goh, 2026).

The individual tools are classical. The new content is the exact source-locked synthesis: the minimal common strip, the divisibility Gram formula, the isolated fourth frequency, the non-arithmetic adversaries, and the full positive-metric collapse theorem.

3 The exact Schatten-3 threshold

Theorem 3.1 (Similarity criterion)**.** *For every $q \geq 1$,*

$$T_s \in \mathcal{S}_q \iff \sum_{p \in A} p^{-q\Re s} < \infty.$$

For the prime atom set this is equivalent to $q\Re s > 1$.

Proof. The identities

$$T_s = Z D_s^A Z^{-1}, \qquad D_s^A = Z^{-1} T_s Z$$

and the two-sided ideal property show that $T_s \in \mathcal{S}_q$ iff $D_s^A \in \mathcal{S}_q$. The latter is diagonal with singular values $p^{-\Re s}$. For primes, the corresponding series converges above exponent one, diverges at exponent one by Euler's prime harmonic theorem, and also diverges below it. □

Theorem 3.2 (Common reflected strip)**.** *For every $q \geq 1$,*

$$\mathcal{B}_s \in \mathcal{S}_q \iff \frac{1}{q} < \Re s < 1 - \frac{1}{q}.$$

The strip is nonempty exactly for $q > 2$. In particular,

$$\mathcal{B}_{1/2+it} \in \bigcap_{q>2} \mathcal{S}_q, \qquad \mathcal{B}_{1/2+it} \notin \mathcal{S}_2.$$

Proof. For $B = \left(\begin{smallmatrix} 0 & A \\ C & 0 \end{smallmatrix}\right)$, $B^*B = \operatorname{diag}(C^*C, A^*A)$, so $B \in \mathcal{S}_q$ iff both legs belong to $\mathcal{S}_q$. Theorem 3.1 gives $q\Re s > 1$ for the first leg and $q(1 - \Re s) > 1$ for the reflected leg. Solving yields the stated strip. □

Proposition 3.3 (Critical-line self-adjointness)**.** *For every real t, $\mathcal{B}_{1/2+it}$ is compact self-adjoint.*

Proof. The incidence idempotents are source-real, and $1 - (1/2 + it) = 1/2 - it = \overline{1/2 + it}$. Therefore $T_{1-s}^\sharp = T_s^*$ on the line. Compactness follows from theorem 3.2, for example with $q = 3$. □

Figure 1 makes the boundary obstruction explicit. The critical-line family is not Hilbert–Schmidt, so its second power is not trace class. This fact determines which spectral moment and which modified determinant are admissible.

Remark 3.4 (Family versus fixed operator). Proposition 3.3 does not construct a single operator with t as spectral variable. It constructs a map $t \mapsto \mathcal{B}_{1/2+it}$ whose value is self-adjoint for each t. This is a positive analytic statement but fails the strict fixed- operator route gate.

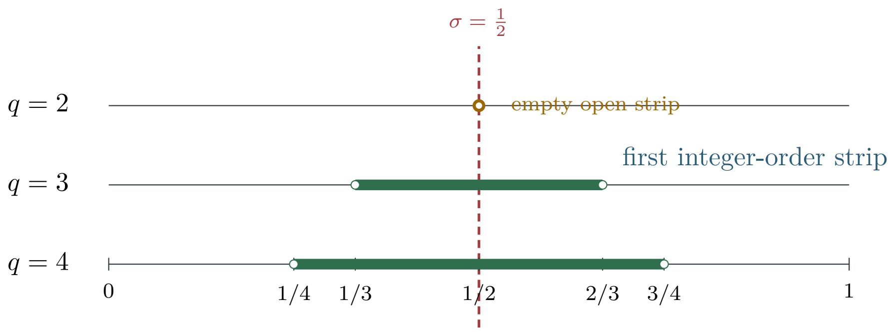

Figure 1: The common domain is $1/q < \sigma < 1 - 1/q$. At $q = 2$ the endpoints meet at the excluded critical point; $q = 3$ is the first integer order giving an open strip through $\sigma = 1/2$.

4 Native mixed-Gram geometry

The source formula for a primitive incidence idempotent is

$$q_n(a, b) = \mathbf{1}_{a|n}\, \mu(b/n)\mathbf{1}_{n|b}.$$

Write $q_n = u_n v_n^{\mathsf{T}}$ and define

$$G_{pq} = \mathrm{Tr}(q_p q_q^{\sharp}), \qquad C_\eta = \sum_{k\ge 1} \frac{\mu(k)^2}{k^{2\eta}} = \frac{\zeta(2\eta)}{\zeta(4\eta)}.$$

Theorem 4.1 (Divisibility Gram coefficients)**.** *For primes p, q,*

$$G_{pp} = C_\eta(1 + p^{-2\eta}),$$

and, if $p \ne q$,

$$G_{pq} = C_\eta \frac{(pq)^{-2\eta}}{(1 + p^{-2\eta})(1 + q^{-2\eta})} > 0.$$

Proof. The weighted rank-one contraction factors as

$$G_{pq} = \left(\sum_a u_p(a) u_q(a) a^{2\eta}\right)\left(\sum_b v_p(b) v_q(b) b^{-2\eta}\right).$$

For $p = q$ the two factors are $1 + p^{2\eta}$ and $p^{-2\eta}C_\eta$. For $p \ne q$, the primal overlap is the shared bottom coordinate 1. A nonzero dual term has $b = pqk$, with k squarefree and coprime to pq; the two Möbius signs multiply to $+1$. Removing the p, q Euler factors from C_η gives the formula. Details are recorded in Appendix A.1. □

Proposition 4.2 (Finite two/three-atom phase)**.** *For a finite atom set F and $s = 1/2 + it$,*

$$\mathrm{Tr}\,\mathcal{B}_s^2 = 2\sum_{p\in F} \frac{G_{pp}}{p} + 4 \sum_{\substack{p<q\\ p,q\in F}} \frac{G_{pq}}{\sqrt{pq}} \cos\left(t \log \frac{q}{p}\right).$$

Proof. Both diagonal blocks of $\mathcal{B}_s^2$ have the same trace. Expanding one of them gives

$$\mathrm{Tr}\,\mathcal{B}_s^2 = 2 \sum_{p,q\in F} p^{-1/2-it} q^{-1/2+it} G_{pq}.$$

Pair the ordered terms (p, q) and (q, p). □

At $\eta = 2$,

$$C_2 = \frac{105}{\pi^4}, \quad G_{22} = \frac{1785}{16\pi^4}, \quad G_{33} = \frac{2870}{27\pi^4}, \quad G_{23} = \frac{105}{1394\pi^4}.$$

The exact two-atom cutoff is therefore

$$G_{22} + \frac{2G_{33}}{3} + \frac{4G_{23}}{\sqrt{6}} \cos\left(t \log \frac{3}{2}\right).$$

For atoms $2, 3, 5$, add the three pair frequencies, with

$$G_{55} = \frac{105 \cdot 626}{625\pi^4}, \quad G_{25} = \frac{105}{10642\pi^4}, \quad G_{35} = \frac{105}{51332\pi^4}.$$

4.1 The second-trace firewall

Proposition 4.3 (No countable second trace)**.** *On the critical line, $\mathcal{B}_s^2$ is not trace class. Its diagonal cutoff contribution satisfies*

$$\sum_p \frac{2G_{pp}}{p} \geq 2C_\eta \sum_p \frac{1}{p} = \infty.$$

Thus proposition 4.2 is an exact finite-cutoff diagnostic, not a conditionally interpreted infinite trace. The mixed terms decay rapidly for $\eta > 1$, but their summability does not repair the missing full trace. The correct countable object begins with third regularization.

5 Third regularization and surviving spectral motion

On the strip $1/3 < \Re s < 2/3$, define the honest modified determinant

$$\det_3(I - z\mathcal{B}_s).$$

For small z its logarithm is

$$\log \det_3(I - z\mathcal{B}_s) = -\sum_{m \geq 3} \frac{z^m}{m} \operatorname{Tr}(\mathcal{B}_s^m).$$

The regularization deletes powers one and two. Odd powers of an off-diagonal block remain off diagonal, so every odd trace vanishes. Consequently

$$\log \det_3(I - z\mathcal{B}_s) = -\frac{z^4}{4} \operatorname{Tr}(\mathcal{B}_s^4) + O(z^6).$$

The second-trace Gram ledger is removed together with its harmonic divergence; the fourth moment is the first legal signal.

Theorem 5.1 (Isolated fourth-moment frequency)**.** *For every pair of distinct primes p, q, the coefficient of*

$$\cos\left(2t \log \frac{q}{p}\right)$$

in $\operatorname{Tr} \mathcal{B}_{1/2+it}^4$ is

$$\frac{4G_{pq}^2}{pq} > 0.$$

Hence the native compact self-adjoint spectrum is nonconstant in t, and $\det_3(I - z\mathcal{B}_{1/2+it})$ is nonconstant in t as an analytic germ in z.

Proof. Put $a_p = p^{-1/2-it}$. On the critical line,

$$\operatorname{Tr} \mathcal{B}_s^4 = 2 \operatorname{Tr}\big((T_s T_s^*)^2\big).$$

The tuple (p, q, p, q) in the expansion contributes

$$2a_p^2 \overline{a_q}^2 \operatorname{Tr}\big((q_p q_q^*)^2\big).$$

Since $q_p q_q^*$ has rank at most one,

$$\operatorname{Tr}\big((q_p q_q^*)^2\big) = \operatorname{Tr}(q_p q_q^*)^2 = G_{pq}^2.$$

Adding the conjugate tuple gives the stated cosine coefficient.

No other tuple can cancel it: a quotient of two products of two primes equals q^2/p^2 only when unique factorization forces the numerator multiset to be $\{q, q\}$ and the denominator multiset to be $\{p, p\}$. Prime cutoffs converge uniformly in $\mathcal{S}_4$ for all real t; continuity of the fourth trace moment passes the isolated coefficient to the countable limit. See Appendix A.2 for the continuity estimate. □

For $(p, q, \eta) = (2, 3, 2)$, the isolated coefficient is exactly

$$\frac{3675}{971618\pi^8} \cos\left(2t \log \frac{3}{2}\right).$$

Remark 5.2 (What has and has not survived). Theorem 5.1 proves genuine spectral motion, not merely motion of eigenvectors. It does not restore the original Euler trace ledger: the determinant is third-regularized, its first visible term is fourth order, and its coefficients involve mixed Gram cycles. The adversaries in Section 7 show that this surviving motion is not arithmetically selective.

6 Positive metric rigidity and atom collapse

Could another positive metric remove the mixed Gram defect while retaining a coupled spectrum? The answer is no for the entire bounded positive class.

Theorem 6.1 (Common positive metrics)**.** *Let G be bounded, positive, and boundedly invertible. Then*

$$Gq_n = q_n^* G \quad \text{for every } n$$

if and only if

$$Z^* G Z = D$$

for a positive bounded diagonal D with bounded inverse. If the condition is imposed only for active atoms, Z^*GZ is diagonal on the active coordinates, has no active–dormant coupling, and may have an arbitrary positive block on the dormant complement.

Proof. With $K = Z^*GZ$, conjugating $Gq_n = q_n^* G$ gives

$$KE_n = E_n K.$$

The joint commutant of all coordinate projections is diagonal. Commutation with only the active E_p annihilates the off-diagonal row and column at every active coordinate, leaving one arbitrary dormant block. Positivity and bounded invertibility are preserved under bounded congruence. □

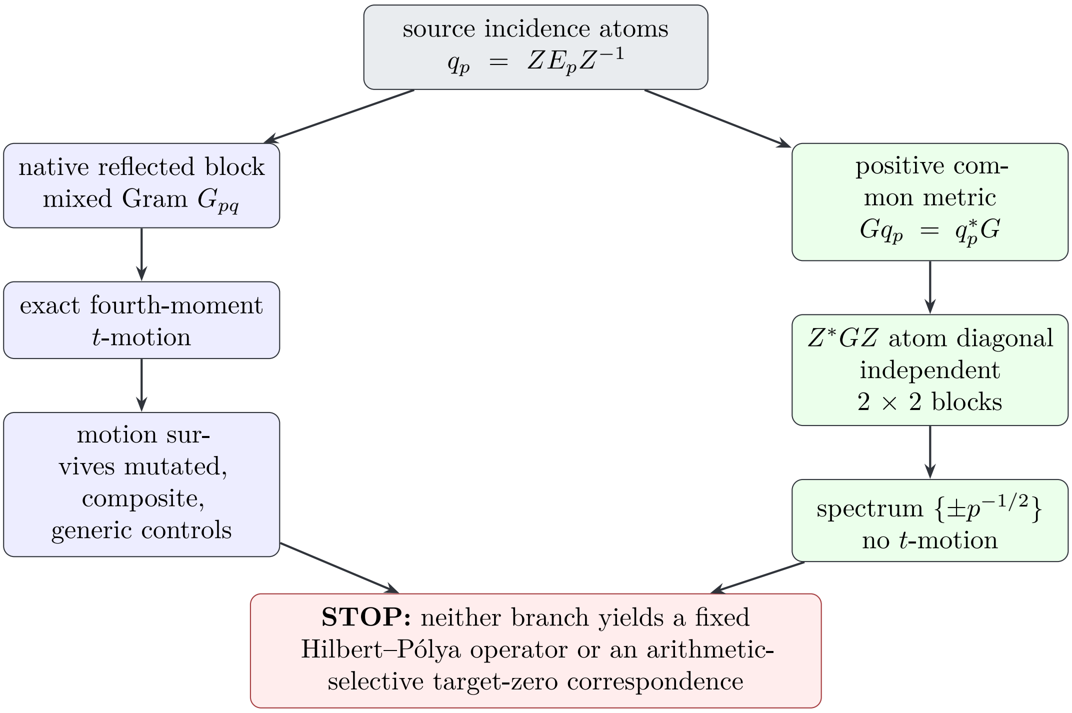

Figure 2: The canonical completion trilemma. The native metric keeps exact spectral motion but generic controls reproduce it. Every positive common metric removes the Gram defect only by forcing independent atom blocks and phase-free spectrum.

Corollary 6.2 (Hellinger/Löwdin collapse)**.** *Let $S = G^{1/2}$, $K = Z^*GZ$, and*

$$U = SZK^{-1/2}.$$

Then U is unitary and

$$ST_sS^{-1} = UD_s^AU^*.$$

Thus every positive common metric that makes the active idempotents self-adjoint sends them to mutually orthogonal coordinate atoms.

Proof. The identity $U^*U = I$ is immediate. By theorem 6.1, K commutes with D_s^A, so the displayed conjugacy follows after inserting $K^{1/2}K^{-1/2}$. □

In the orthogonal atom basis, the reflected block is the direct sum of

$$\begin{pmatrix} 0 & p^{-s} \\ p^{-(1-s)} & 0 \end{pmatrix}.$$

Each block squares to $p^{-1}I_2$, so its paired eigenvalues are $\pm p^{-1/2}$, independent of s. The third-regularized product is

$$\det_3(I - z\widehat{\mathcal{B}}_s) = \prod_p \left(1 - \frac{z^2}{p}\right) \exp\left(\frac{z^2}{p}\right).$$

It converges because each logarithm is $O(p^{-2})$, but it is independent of t. Its auxiliary zeros $z = \pm\sqrt{p}$ are explicit atom values and have no target-zero interpretation.

Figure 2 is a scoped no-go. It classifies bounded positive common metrics; it does not classify indefinite or unbounded metric operators, whose domains and determinant theory would constitute a different problem.

Fixture	active labels	mixed Gram	native B^2	native B^4	metric phase
standard divisibility, $N = 30$	$2, 3, 5$	yes	moves	moves	absent
mutated divisibility, $N = 12$	$2, 3, 5, 6$	yes	moves	moves	absent
composite-only subposet	$4, 6, 9$	yes	moves	moves	absent
seeded locally finite DAG	$10, 14, 21$	yes	moves	moves	absent

Table 1: Aggregate adversaries. The native phase survives every fixture, while every positive-metric characteristic polynomial is phase independent.

fixture	mixed Gram	native B^2	native B^4	metric phase
standard divisibility	nonzero	moves	moves	absent
mutated divisibility	nonzero	moves	moves	absent
composite-only poset	nonzero	moves	moves	absent
seeded generic DAG	nonzero	moves	moves	absent

Figure 3: Every source fixture has native mixed-Gram and fourth-order motion. Orthogonalization erases the phase in every fixture. The matrix visualizes lack of arithmetic selectivity, not relative effect size.

7 Deterministic exact adversarial audit

The exact prototype compiles every idempotent from a finite poset relation as $q_x = ZE_xZ^{-1}$. It does not supply a projector table or a primality oracle. All identities use integers, rational arithmetic, or exact symbolic radicals; floating point is used only to display common-t samples.

The mutated relation deletes $2 < 6$ and $3 < 6$, making 6 an atom. The composite-only subposet has bottom covers $4, 6, 9$. The generic fixture is a deterministic DAG with seed 2801. Exact witnesses include

$$G_{23}^{(N=30)} = \frac{313}{405000}, \qquad G_{4,6}^{(\mathrm{comp})} = \frac{1}{20736}, \qquad G_{10,14}^{(\mathrm{DAG})} = \frac{1}{390625}.$$

These nonzero rational entries establish that the phase signal is not rounding noise.

7.1 Representative common-t samples

Fixture	$\mathrm{Tr}\, B^2(0)$	$\mathrm{Tr}\, B^2(1.3)$	$\mathrm{Tr}\, B^4(0)$	$\mathrm{Tr}\, B^4(1.3)$
standard	2.3055980793	2.3053415717	1.0173068797	1.0168511063
mutated	2.5967636512	2.5966667902	1.0287302339	1.0285865294
composite-only	1.1031574816	1.1031514332	0.2215897527	0.2215846316
seeded DAG	0.4773464242	0.4773463427	0.0400704387	0.0400704092

Table 2: Displayed evaluations of exact symbolic expressions. The proof of nonconstancy uses nonzero exact Laurent coefficients, not the decimal differences.

7.2 Reproducibility and inference boundary

The frozen prototype has SHA-256

e29553c5a04cb31393b6ef8f93d2718285bac52d956cffc04d4c8d53fc6cc737,

and the result JSON has SHA-256

118ae2e85e4ce8d403673f1d00725520c7137a3a032bb9201cda186a61cb5cfb.

Two fresh executions were byte-identical and every exact control passed. The computation certifies finite formulas, control behavior, and metric identities. It does not replace the countable Schatten, Fourier-isolation, or metric-rigidity proofs.

8 The completion trilemma and strict route

The two canonical completions now admit a direct comparison.

Property	Native reflected metric	Positive common metric
source incidence ownership	retained	retained under metric transfer
critical self-adjointness	yes, as a t-dependent family	yes
native mixed Gram	retained	removed
fourth-order t-motion	exact and nonzero	erased
arithmetic selectivity	fails aggregate controls	atom inventory only
Euler determinant identity	not retained by chiral $\det_3$	phase-free atom product
fixed spectral operator	no	no

Table 3: The canonical completion trilemma. Neither column retains source ownership, selective critical motion, and orthogonal purity simultaneously.

The native construction proves more than a formal analogy: it gives an analytic modified determinant and a compact self-adjoint spectrum whose fourth moment moves. Yet Table 1 shows that the motion is generic. The metric construction removes this ambiguity only by reducing the active sector to independent coordinate atoms.

8.1 Strict gate evaluation

Gate	Status	Reason
A0	structural arithmetic relation	The compiler is derived from divisibility incidence and Möbius inversion.
A1	`FAIL`	The chiral completion does not preserve a pure Euler word/orbit ledger.
A2	analytic determinant	$\det_3(I - z\mathcal{B}_s)$ is honest on $1/3 < \Re s < 2/3$.
A3	`FAIL`	Self-adjointness holds for a t-dependent family, not one fixed operator.
A4	`FAIL`	No continuation or target-zero equivalence theorem exists.

Table 4: Frozen strict route evaluation for SD-C30.

Frozen decision.

(`A0_STRUCTURAL_ARITHMETIC_RELATION`, `A1_FAIL`,
`A2_ANALYTIC_DETERMINANT`, `A3_FAIL`, `A4_FAIL`)

`ROUTE_A_REJECTED`. Route B is locked false. Target-zero fields are `NA`.

No target-zero coordinates, fitting objective, or zero-labeled statistic appears in the construction or audit. The auxiliary zeros of the metric product are stated only as zeros in its z variable.

9 Conclusion

The reflected incidence construction clears a real analytic barrier. Its common Schatten domain is exact, the first integer order is $\mathcal{S}_3$, and on the critical line it gives a compact self-adjoint family. The fourth moment contains an isolated positive frequency, so native spectral motion is a theorem rather than a numerical impression.

That gain sharpens the obstruction. The exact motion is reproduced by mutated, composite-only, and generic locally finite posets. It comes from mixed Gram geometry, not from a demonstrated arithmetic selection rule. Every bounded positive metric that makes the active incidence idempotents self-adjoint is forced into their coordinate commutant. Its Hellinger/Löwdin transfer produces independent atom blocks and an s-independent modified product.

The resulting contribution is a canonical completion trilemma. The native metric retains motion but not selectivity; the positive metric retains orthogonality but not motion; neither gives a fixed Hilbert–Pólya operator. This is a strict route rejection, but it also identifies a focused next problem: classify source-natural renormalizations that subtract the diagonal prime-harmonic divergence without retaining a generic mixed-poset signal. A successful next step must distinguish global divisibility coherence, not merely pairwise obliqueness.

References

Jorge Antezana, Gustavo Corach, Mariano Ruiz, and Demetrio Stojanoff. Weighted projections and Riesz frames, 2004. URL https://arxiv.org/abs/math/0411326.

Thomas Britz, Alan Carey, Fritz Gesztesy, Roger Nichols, Fedor Sukochev, and Dmitriy Zanin. The product formula for regularized fredholm determinants, 2020. URL https://arxiv.org/abs/2007.12834.

Marcel Goh. An uncertainty principle for möbius inversion on posets. *Contributions to Discrete Mathematics*, 21(1), 2026. doi: 10.55016/6ej3y980. URL https://doi.org/10.55016/6ej3y980.

Nikolaos Koutsonikos-Kouloumpis and Matthias Lesch. The product formula for regularized fredholm determinants: Two new proofs, 2022. URL https://arxiv.org/abs/2202.12923.

Per-Olov Löwdin. On the non-orthogonality problem connected with the use of atomic wave functions in the theory of molecules and crystals. *The Journal of Chemical Physics*, 18(3): 365–375, 1950. doi: 10.1063/1.1747632.

Gian-Carlo Rota. On the foundations of combinatorial theory i. theory of möbius functions. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, 2(4):340–368, 1964. doi: 10.1007/BF00531932.

Barry Simon. *Trace Ideals and Their Applications*, volume 120 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, second edition, 2005. doi: 10.1090/surv/120.

Wai-Shing Tang. Oblique projections, biorthogonal Riesz bases and multiwavelets in Hilbert spaces. *Proceedings of the American Mathematical Society*, 128(2):463–473, 2000. doi: 10.1090/S0002-9939-99-05075-3.

A Additional proof details

A.1 Weighted rank-one contraction

In unnormalized coordinates the weighted reflection is $X^\sharp = W_\eta^{-1} X^\top W_\eta$. For $q_p = u_p v_p^\top$,

$$q_q^\sharp = W_\eta^{-1} v_q u_q^\top W_\eta.$$

Consequently

$$\begin{aligned}\operatorname{Tr}(q_p q_q^\sharp) &= \operatorname{Tr}\left(u_p v_p^\top W_\eta^{-1} v_q u_q^\top W_\eta\right)\\ &= (u_q^\top W_\eta u_p)(v_p^\top W_\eta^{-1} v_q).\end{aligned}$$

This also makes symmetry and nonnegativity of the divisibility coefficients transparent.

For distinct primes, write $b = pqk$. If k contains p or q, one of $\mu(b/p), \mu(b/q)$ vanishes. If any other prime divides k twice, both vanish. Otherwise

$$\mu(qk)\mu(pk) = \mu(k)^2 = 1.$$

Thus

$$\begin{aligned}v_p^\top W_\eta^{-1} v_q &= (pq)^{-2\eta} \sum_{\substack{k \text{ squarefree} \\ (k,pq)=1}} k^{-2\eta}\\ &= (pq)^{-2\eta} \frac{C_\eta}{(1+p^{-2\eta})(1+q^{-2\eta})}.\end{aligned}$$

A.2 Uniform fourth-moment passage

Let F_N be increasing finite prime sets and

$$T_{s,N} = Z\left(\sum_{p\in F_N} p^{-s} E_p\right) Z^{-1}.$$

On the critical line,

$$\sup_{t\in\mathbb{R}} \|T_{1/2+it} - T_{1/2+it,N}\|_4 \le \|Z\|\,\|Z^{-1}\| \left(\sum_{p\notin F_N} p^{-2}\right)^{1/4} \longrightarrow 0.$$

The same estimate holds for the reflected leg, hence for $\mathcal{B}$. For $A, B \in \mathcal{S}_4$, the telescoping identity and Hölder's trace inequality give, on norm-bounded sets,

$$|\operatorname{Tr}(A^4) - \operatorname{Tr}(B^4)| \le C\|A-B\|_4.$$

Therefore the finite trigonometric polynomials $t \mapsto \operatorname{Tr}\mathcal{B}^4_{1/2+it,N}$ converge uniformly. Their Bohr Fourier coefficients pass to the limit.

To isolate frequency $2\log(q/p)$, a term with indices (a,b,c,d) would require

$$\frac{bd}{ac} = \frac{q^2}{p^2}.$$

Both sides are quotients of products of two primes. Equality of prime exponent vectors forces $a = c = p$ and $b = d = q$. Thus the coefficient computed in theorem 5.1 is unique.

A.3 Active commutant

Suppose $KE_p = E_pK$ for an active coordinate p. For $i \neq p$, the (i,p) entry of the left side is K_{ip} and of the right side is zero, so $K_{ip} = 0$. The (p,i) entry similarly gives $K_{pi} = 0$. Applying this to each active p proves the active one-dimensional blocks in theorem 6.1.

If K is positive and boundedly invertible, functional calculus defines $K^{-1/2}$. Then

$$U^*U = K^{-1/2}Z^*GZK^{-1/2} = I.$$

Since K is block diagonal with scalar active blocks, it commutes with D_s^A. This completes the unitary-collapse calculation without assuming that the dormant block is diagonal.

A.4 Modified atom product

For one orthogonalized atom, the chiral block has eigenvalues $\lambda_\pm = \pm p^{-1/2}$. The third modified factors for $I - zB$ are

$$(1 - z\lambda_\pm)\exp\left(z\lambda_\pm + \frac{z^2\lambda_\pm^2}{2}\right).$$

Multiplying the pair cancels the linear exponentials and yields

$$\left(1 - \frac{z^2}{p}\right)e^{z^2/p}.$$

Since

$$\log(1 - z^2/p) + z^2/p = O(p^{-2})$$

locally uniformly in z, the prime product converges normally.

B Scope and ownership declarations

1. **Arithmetic specialization.** Every critical-line theorem uses $u = 1$. If $T_s(u) = \sum_p u^{\ell(p)}p^{-s}q_p$ is retained, an r-fold atom contribution carries $u^{r\ell(p)}$. The altered threshold for $|u| < 1$ is not continuation of the $u = 1$ result.
2. **Trace ownership.** The finite formula for $\operatorname{Tr}\mathcal{B}_s^2$ is a cutoff identity. The countable $\mathcal{B}_{1/2+it}$ is not Hilbert–Schmidt, so no ordinary second trace is asserted.
3. **Determinant ownership.** The Euler determinant of T_s on $\Re s > 1$ and the chiral $\det_3(I - z\mathcal{B}_s)$ on $1/3 < \Re s < 2/3$ are different objects. The latter deletes powers one and two.
4. **Spectral ownership.** The compact self-adjoint operator depends on t. No fixed operator with spectral parameter t is claimed.
5. **Metric scope.** Metric rigidity covers bounded positive boundedly invertible common metrics. It does not classify indefinite or unbounded metrics.
6. **Control inference.** Mutated, composite-only, and generic-poset controls reproduce the native motion. Their role is to reject arithmetic selectivity.
7. **Target-zero firewall.** No target-zero position, label, fitting criterion, or comparison enters the definitions, the exact audit, or the route decision. Auxiliary zeros in z are not identified with zeta zeros.
8. **Route status.** The frozen tuple is

   (`A0_STRUCTURAL_ARITHMETIC_RELATION`, `A1_FAIL`,
   `A2_ANALYTIC_DETERMINANT`, `A3_FAIL`, `A4_FAIL`).

   Overall status is `ROUTE_A_REJECTED`; Route B is locked.